

Shielding firm value: Employment protection and process innovation*

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Abstract

We show that an increase in labor dismissal costs leads firms to increase their process innovation, especially in industries with a large share of labor costs. Firms with a greater stock of knowledge capital adjust their production methods and mitigate the effects of increased labor rigidity. They exhibit larger increases in process innovation, larger decreases in employment and employment growth rates, and larger increases in capital intensity. These adjustments allow them to increase labor productivity, operating performance, and ultimately to avoid value losses.

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1. Introduction

Frictions in labor markets that make labor rigid and costly to adjust lead to higher production costs and higher operating risks for firms. Such frictions thus hurt firm performance and value. To mitigate the impact of these frictions, firms reduce their financial leverage (Simintzi, Vig, and Volpin (2015), Serfling (2016)) or alter their production methods by substituting capital for labor and outsourcing production (Autor (2003), Autor, Kerr, and Kugler (2007)). Adjustments in production methods often require inventions of new ways to produce. For example, to take advantage of a higher capital-labor ratio a firm may develop new production methods that are appropriate for that capital-labor ratio (Jones (2015)). The ability to invent new production methods is thus critical in firms' responses to labor market shocks, yet little is known about this role of innovation empirically.

In this paper, we study how innovation helps firms respond to labor market rigidities, and to this end we focus specifically on an increase in labor dismissal costs. By increasing the cost of future layoffs, labor dismissal costs reduce the efficiency of firms' hiring and firing decisions, that is, firms retain workers with wages above their productivity and refrain from hiring workers with wages below their productivity. Labor dismissal costs thus create operating leverage and effectively increase the cost of labor relative to other production inputs, which can reduce a firm's value both by decreasing its expected future cash flows and by increasing its cost of capital. Our hypothesis is that the introduction of such labor rigidities leads firms to invent new production methods that introduce cost-saving technologies and allow firms to substitute capital for labor.¹

To test this hypothesis, we create measures of innovation that distinguish *process innovations*, which refer to inventions of new methods used to adjust firms' production (Scherer (1982, 1984), Eswaran and Gallini (1996)) from other non-process innovations. The basis for our novel measures is the textual analysis of patent claims. Claims are written using stilted legalistic language and consistent vocabulary across firms and over time which allows us to accurately distinguish patent claims that describe process innovations from those that do not. Our main measures thus count the number of process and non-process claims contained in patents filed by each firm. To account for the significant heterogeneity in patent quality, we also use citation-weighted counts of process and non-process patents.

¹ This intuition dates back to John Hicks (1932), who noted that “...a change in the relative prices of the factors of production is itself a spur to invention, and to invention of a particular kind—directed to economizing the use of a factor which has become relatively expensive...”

We exploit variation in labor dismissal costs associated with the staggered adoption of the “good faith” exception to the common law “employment at will” doctrine by U.S. state courts between 1973 and 1995. This doctrine gives employers unlimited discretion to fire employees at any time. The good faith exception significantly restricts this discretion because it serves as a general prohibition against firing workers without just cause and it thus opens firms to potentially costly litigation if they lay off workers (Dertouzos and Karoly (1992), Kugler and Saint-Paul (2004), Autor, Kerr, and Kugler (2007)). Prior work also shows that the adoption of this exception leads to a significant increase in labor adjustment costs and operating leverage (Autor, Kerr, and Kugler (2007), Serfling (2016)).

We find that, following the adoption of the good-faith exception, firms located in adopting states increase their process innovation by 6.1% to 8.9% relative to firms in non-adopting states. This increase in process innovation becomes statistically and economically significant three years after the adoption. Further, the increase in process innovation is more pronounced in industries where labor costs account for a larger share of production costs, and thus are more impacted by the law. The adoption of the good-faith exception does not, however, materially affect non-process innovation. These results suggest that higher labor adjustment costs lead firms to develop new production methods that lower their costs.

The evidence supports a causal interpretation of our results. First, the good-faith exception concerns firms’ firing decisions rather than innovation outcomes and its recognition follows from court decisions made by independent judges in specific cases rather than from changes in state legislation that are potentially contentious. Second, in addition to firm and year fixed effects, our results hold controlling for various time-varying firm characteristics, as well as controlling for the state’s business environment and political leaning. When possible, we also include state fixed effects interacted with year fixed effects, which ensures that state-specific shocks cannot explain our findings. Third, the pre-treatment trends in the innovation of firms in treated and control states are indistinguishable. Fourth, the adoption of the good-faith exception in neighboring states does not affect a firm’s innovation, suggesting that local economic shocks correlated with the adoption of the good-faith exception are unlikely to drive our results. Fifth, our results do not seem to be driven by a differential impact across states of increased patent enforcement in the early 1980s and associated surge in patenting activity (Lerner and Seru (2017)).

The adoption of the good-faith exception in a firm’s state of headquarters can also increase the job security of a firm’s inventors employed in that state and, through this

mechanism, lead them to exert more effort in innovation (Acharya, Baghai, and Subramanian (2014)). If such effort is focused only on processes, this rationale could explain our results. Employment contracts are typically governed by labor laws in the state where the employee works, so the adoption of the good-faith exception in a firm's state of headquarters does not affect the job security of inventors the firm employs in other states. Inconsistent with the job security channel, we find that the adoption of the good-faith exception in a firm's state leads to an increase in the firm's process innovation even if the majority of its lead inventors are located in other states or if the lead inventors are dispersed across several states. We continue to find no effect on non-process innovation irrespective of the location of inventors.

Importantly, we show that firms with a greater stock of knowledge capital (measured by their stock of prior patents) respond to increased labor dismissal costs by developing considerably more process innovations. In contrast, non-process innovation remains unaffected even for firms with a large stock of knowledge capital. Furthermore, consistent with our intuition that innovation helps mitigate the negative effects of labor market rigidities on firms, we also find that firms with a greater stock of knowledge capital reduce their total employment and employment growth and increase their capital intensity. Such adjustments in production input mix are insignificant for firms with a smaller stock of knowledge capital. These results suggest that, when an increase in labor rigidity makes the labor input more costly, a greater stock of knowledge capital facilitates the creation of process innovations which firms then deploy as new production methods.

Such firm responses help mitigate the adverse effects of labor market rigidities on firm performance and value. Following the adoption of the good-faith exception, we document increases in the productivity of labor for firms with a greater stock of knowledge capital and decreases for firms with a smaller stock of knowledge capital. We also find that firms with a greater stock of knowledge capital maintain their profitability and avoid market value losses when increased labor rigidity impacts their business. Overall, our evidence highlights that innovation is a key driver of a firm's success in adjusting to changing conditions in input markets and thereby an important driver of firm value.

Our paper advances the literature that examines how frictions in labor markets affect corporate policies and valuations. It is closely related to prior work that shows how employment protection creates operating leverage and, through this mechanism, impacts corporate investment and financing decisions (Autor, Kerr, and Kugler (2007), Simintzi, Vig,

and Volpin (2015), Serfling (2016), Fairhurst and Serfling (2017)).² We show that employment protection leads innovative firms to tilt their R&D toward developing process innovations in order to deploy new production methods that mitigate the impact of increased labor rigidity on firm value.

A few prior studies examine how frictions in labor markets affect innovation. Acharya, Baghai, and Subramanian (2014) show that the adoption of the good-faith exception leads firms to increase their overall patented innovation because increased job security incentivizes firms' inventors to exert more effort. We show that this increase in innovation is largely driven by process innovation. We also highlight a new mechanism through which labor protection affects innovation, namely, the need to invent new processes that allow firms to adjust their production methods and mitigate the negative impact of labor rigidity on their values. Bradley, Kim, and Tian (2016) show that an increase in labor unionization leads to a reduction in innovation, because unions can hold up the firm and demand higher wages once the firm has incurred the sunk cost of innovation, which reduces the firm's ex-ante incentives to innovate. This suggests that different types of labor markets frictions impact firm's innovation differently. Last, Bena and Simintzi (2017) show that large U.S. multinational firms reduced their process innovation after their access to "cheap" offshore labor improved. We instead focus on changes in the effective price of domestic labor that affect all U.S. firms, and show that employment protection stimulates process innovation, allowing firms to reduce their reliance on domestic labor.

A broader corporate finance literature examines what factors stimulate innovation, such as governance (Atanassov (2013), Balsmeier, Fleming, and Manso (2017)), ownership structure (Aghion, Van Reenen, and Zingales (2013), Bena, Ferreira, Matos, and Pires (2017)), organization (Lerner, Sørensen, and Strömberg (2011), Ferreira, Manso, and Silva (2014), Seru (2014)), managerial compensation (Manso (2011), Ederer and Manso (2013)), capital structure (Atanassov, Nanda, and Seru (2007)), litigation risk (Cohen, Gurun, and Kominers (2016)), corporate taxes (Mukherjee, Singh, and Žaldokas (2017)), and the supply of credit (Amore, Schneider, and Žaldokas (2013)). We add to this work by highlighting that the distinction between process and other types of innovation might be useful in advancing the literature on what drives innovation.

² Related work in the labor and finance literature includes, Atanassov and Kim (2009), Lee and Mas (2009), Klasa, Maxwell, and Ortiz-Molina (2009), Matsa (2010), Chen, Kacperczyk, and Ortiz-Molina (2011, 2012), Agrawal and Matsa (2013), and Favilukis and Lin (2016).

The rest of the paper is organized as follows. Section 2 discusses our measures of process and non-process innovation, as well as our identification strategy. Section 3 discusses our empirical results. Section 4 concludes.

2. Innovation measures, wrongful discharge laws, and summary statistics

2.1 Patented process and non-process innovation

Our main dependent variables are designed to separately measure a firm’s patented process and non-process innovation output. Prior literature highlights that process innovation describes a new way to produce an existing good and aims to improve a firm’s own production methods to lower its production costs (Scherer (1982, 1984), Link (1982), Cohen and Klepper (1996), Eswaran and Gallini (1996)).

As in Bena and Simintzi (2017), we first obtain full texts of the universe of utility patents awarded by the United States Patent and Trademark Office (USPTO) to U.S. and international companies, individuals, and other institutions from January 1976 to December 2012. We then parse the structured-texts of patent grants to identify the section that contains patent claims. Claims define – in technical terms – the scope of protection conferred by a patent and thus define what subject matter the patent protects. Since claims are critical defining elements of a patent, they are the primary subject of examination in patent prosecution and crucial in patent litigation cases. Last, we use text analysis techniques to examine patent claims and, aided by the very legalistic and stilted language used in drafting claims, we unambiguously distinguish claims within each patent as either process or non-process (see Bena and Simintzi (2017) for more extensive discussion).

We measure a firm’s annual process and non-process innovation in two ways. The first approach exploits our ability to break down each and every patent filing into individual process and non-process claims it contains. We compute a firm’s *Process Claims* by summing the number of process claims contained in all of its patent filings in each year. Similarly, we compute a firm’s *Non-Process Claims* by summing the number of non-process claims contained in all of its patent filings in each year. We set both variables to zero for firm-years with no patent filings. These measures capture all process and non-process innovation patented by a firm in each year. Counts of claims, however, ignore differences in the quality of innovations protected by each specific claim. This limitation cannot be addressed using measures based on claims, because there are no indicators of quality, for example, the

number of citations received, of individual claims.

Our second approach uses the number of citations received by each patent to account for the differences in the scientific merit of innovations (Hall, Jaffe, and Trajtenberg (2001, 2005), Jaffe and Trajtenberg (2002)). To ensure that we keep a clear distinction between process and non-process innovations, we focus on patents that contain only process claims (process patents) or only non-process claims (non-process patents), but not both.³ *C-W Process Patents* is the citations-weighted number of process patents and *C-W Non-Process Patents* is the citations-weighted number of non-process patents filed by a firm in each year, respectively. We set both measures to zero for firm-years with no process patents and no non-process patents.

To match patent filings to firms in Compustat, we first search each patent grant document and identify the names of patent assignees, the country of these assignees, and whether each assignee is a U.S. corporation, a non-U.S. corporation, an individual, or a government body. Using this information, we then match patents to firms in Compustat. Our matching algorithm involves two main steps. First, we standardize patent assignee names and firm names, focusing on unifying suffices and dampening the non-informative parts of firm names. Second, we apply multiple fuzzy string matching techniques to identify the firm, if any, to which each patent belongs (see Bena, Ferreira, Matos, and Pires (2017) for more extensive discussion).

2.2 Wrongful discharge laws as a source of exogenous variation in labor dismissal costs

We use a difference-in-differences methodology based on plausibly exogenous variation in labor dismissal costs associated with the staggered adoption of “wrongful discharge laws” by U.S. state courts during the late 1970s and the early 1990s. Prior work extensively discusses these laws and their economic significance as a source of exogenous variation in labor dismissal costs (Dertouzos and Karoly (1992), Miles (2000), Kugler and Saint-Paul (2004), Autor, Donohue, and Schwab (2004, 2006), Autor, Kerr, and Kugler (2007)). Recently, Acharya, Baghai, and Subramanian (2014) and Serfling (2016) further validate this setting for identification purposes.

The legal framework regarding worker dismissals prevailing in the United States at the beginning of the 1970s was centered around the common law doctrine of “employment at

³ Process patents and non-process patents account for 70.3% of all patents in our data.

will". This doctrine sustained a legal presumption that employers can fire workers *at will*, that is, "for good cause, bad cause, or no cause" (a quote from *Payne vs. Western & Atlantic Railroad*, Supreme Court of Tennessee, 1884), which essentially gave employers unlimited discretion in dismissing workers. Between 1973 and 1995, courts in 46 states set new common-law legal precedents that recognized three different kinds of exceptions to employment at will usually referred to as "wrongful discharge laws". These exceptions significantly limited employers' discretion to fire workers, opened them to potentially costly litigation, and generated uncertainty about when employers could terminate workers with impunity. The survey of public firms' managers conducted by Dertouzos, Holland, and Ebener (1988) indicated that 46% of managers feared potential losses arising from such lawsuits, while Jung (1997) and Boxold (2008) document economically significant awards to plaintiffs.

The key legal change we use to gauge an increase in labor dismissal costs is the adoption of the "*good-faith exception*" to the employment at will doctrine by state courts. This exception represents an implied covenant to terminate employment only in good faith and fair dealing, which essentially prevents an employer from firing workers for any "bad cause". The good-faith exception represents the largest deviation from at-will employment and is the most far reaching. This exception gives employees both a contract and a tort cause of action in the event they are dismissed, allowing them to seek compensation for both contractual losses and emotional distress, as well as punitive damages that imply highly uncertain amounts. Importantly, it serves as a general prohibition against terminating any worker without just cause (economic necessity or poor performance) and thus could have sweeping consequences (Dertouzos and Karoly (1992), Kugler and Saint-Paul (2004), Autor, Kerr, and Kugler (2007)).

State courts adopted two other exceptions to the employment at will doctrine during this period that had less impact on firms. We include them as control variables in our analyses. One is the "*implied-contract exception*", which protects workers from discharges when there is an implicit promise that the firm will not fire workers without good cause.⁴ The other is the "*public-policy exception*", which protects workers against discharges that would thwart an important public policy, such as, performing jury duty, filing a worker's compensation claim, reporting an employer's wrongdoing, or refusing to commit perjury.⁵

⁴ Such promises follow from the language in employment contracts and personnel manuals, or from expectations of continuing employment based on the length of service and prior promotions. In practice, employers can reword these documents to clearly indicate that employment contracts are at will (Miles (2000), Autor, Kerr, and Kugler (2007)) and employers indeed took such steps (Sutton and Dobbin (1996)).

⁵ This exception is of minor legal and economic significance, because large compensation amounts are rare and

Prior work highlights that the adoption of good-faith exception significantly increases hiring and firing costs, ultimately increasing labor adjustment costs and operating leverage. Using plant-level data from the Census Bureau, Autor, Kerr, and Kugler (2007) show that the adoption of this exception reduces employment volatility, both in the intensive (within-plant) and the extensive (plant entry/exit) margins.⁶ Further, in a sample of public firms, Serfling (2016) shows that it leads to a reduction in the volatility of employment and in the likelihood that a firm fires workers when its earnings decrease; it also leads to an increase in the elasticity of changes in earnings to changes in sales and in the volatility of profits.

The adoption of the good-faith exception is unlikely to be driven by changes in economic or political conditions in the state that could be correlated with firms’ incentives to innovate. It occurs in judiciary decisions by state courts and not through a potentially contentious legislative process in the state. The judges are independent and largely immune to political pressure from interested parties, and thus base their decisions on the merits of the specific case. Further, Walsh and Schwarz (1996) document three key reasons why judges adopt these exceptions: enhancing fairness in employment relationships, assuring consistency with established principles of contract law, and following similar rulings in other states. Last, Acharya, Baghai, and Subramanian (2014) and Serfling (2016) show that the adoption of the good-faith exception is not driven by changing political conditions, economic conditions, or in other labor regulation in the state, and that such rulings were not anticipated. Hence, it provides plausibly exogenous variation in employment protection in our study of innovation.

2.3 Data, sample, and summary statistics

The sample spans the period 1975-1997 and is based on the publicly traded firms in the Compustat dataset. As in Bloom, Schankerman, and Van Reenen (2013), it includes all non-financial and non-utility firms headquartered in the U.S. that filed at least one patent with the United States Patent and Trademark Office (USPTO) during this period. To merge the wrongful discharge laws indicators and other state-level variables into our dataset, we use a firm’s *historical* state of headquarters obtained from the “company header history” file in the

because courts typically limit public policy cases to clear violations of specific legislative commands rather than violations of a vaguer sense of public obligation (Edelman, Abraham, and Erlanger (1992), Autor, Donohue, and Schwab (2006)).

⁶ Also consistent with these legal changes affecting hiring decisions, Kugler and Saint-Paul (2004) further show that the good-faith exception reduced the re-employment probability of unemployed relative to employed workers.

legacy CRSP/Compustat Merged database (see the Online Appendix for more details).⁷

The Appendix contains the definitions of all variables we use in our analyses and Table 1 reports the summary statistics for our sample of 4,457 innovative firms. Regarding our key innovation variables, the mean $\text{Log}(1 + \text{Process Claims})$ is 1.133 and the mean $\text{Log}(1 + \text{Non-Process Claims})$ is 1.872. The median value is zero for both variables and the value associated with the 90th percentile is 4.025 for $\text{Log}(1 + \text{Process Claims})$ and 5.017 for $\text{Log}(1 + \text{Non-Process Claims})$. Hence, while there are more non-process than process claims in the patents contained in our data, both variables exhibit similar distributions which, in turn, closely resemble the typical distribution of patent count variables reported in prior studies.

3. Results

3.1 Difference-in-differences specification

To identify the effect of wrongful discharge laws on process and non-process innovation, we estimate the following difference-in-differences regression:

$$y_{i,s,t} = \beta_1 \text{Good Faith}_{s,t} + \beta_2 \text{Implied Contract}_{s,t} + \beta_3 \text{Public Policy}_{s,t} + \delta X_{i,s,t-1} + \alpha_i + \mu_t + \varepsilon_{i,s,t}$$

where i denotes firm, s denotes the firm's state of headquarters, and t denotes year. The dependent variables y are $\text{Log}(1 + \text{Process Claims})$, $\text{Log}(1 + \text{Non-Process Claims})$, $\text{Log}(1 + \text{C-W Process Patents})$, and $\text{Log}(1 + \text{C-W Non-Process Patents})$. The three indicator variables for whether the state recognizes the corresponding exception to the employment at will doctrine are *Good Faith*, *Implied Contract*, and *Public Policy*, respectively. The vector X includes lagged firm-level control variables ($\text{Log}(1 + \text{Patent Stock})$, $\text{Log}(1 + \text{R\&D Stock})$, $\text{Log}(\text{Sales})$, and $\text{Log}(M/B)$) and state-level control variables (*State GDP Growth* and *State Political Balance*). As in Henderson and Cockburn (1996) and Bloom, Schankerman, and Van Reenen (2013), we control for differences in a firm's stock of knowledge using lagged patent stock and lagged R&D stock, as well as for firm size and investment opportunities. Further, the inclusion of our state-level control variables alleviates concerns that changes in a state's economic conditions or political environment correlated with both the adoption of wrongful discharge legislation and innovation could confound our inferences. Last, α_i is a firm fixed

⁷ According to employment law, the relevant jurisdiction for a wrongful discharge lawsuit is the state where the employee works. Firms often employ workers in different states, but data restrictions only allow us to identify a firm's state of headquarters, where most of a firm's workers are typically employed.

effect, which controls for firm characteristics that do not vary over our sample period, and μ_t is a year fixed effect, which absorbs time-varying shocks affecting all firms. We cluster standard errors by state.

The staggered adoption of wrongful discharge laws allows a time-varying control group that provides a counterfactual for how firms' innovations would have evolved in the treated states had they not adopted such legislation. The key coefficient of interest is β_1 , which gauges the estimate of the adoption of the good-faith exception on innovation. This coefficient captures the change in innovation for firms in adopting (treated) states relative to the contemporaneous change in innovation for firms in non-adopting (control) states.

3.2 Impact of labor dismissal costs on process vs. non-process innovation

Table 2 reports the results of pooled (panel) OLS regressions that implement our main difference-in-differences approach discussed in Section 3.1. We report specifications with and without control variables. The inclusion of control variables helps control for differences between firms in treatment and control states that could confound our inferences given the lack of truly random assignment to these groups. However, it is conceivable that some of these variables could be affected by the adoption of wrongful discharge laws and bias our estimates. Hence, both specifications are useful in assessing the causal effect of wrongful discharge laws on innovation.

In Panel A, the dependent variables are our measures of process innovation $\text{Log}(1 + \text{Process Claims})$ in columns (1)-(3) and $\text{Log}(1 + \text{C-W Process Patents})$ in columns (4)-(6). Columns (1)-(2) and (4)-(5) show that the adoption of the good-faith exception leads to a statistically significant increase in the process innovation of firms in adopting states relative to that of firms in non-adopting states both in specifications with and without control variables. This effect is also economically significant: Process innovation increases by 13.4% when measured by process claims (column (1)) and it increases by 9.1% when measured by citations-weighted process patents (column (4)). The magnitudes decrease to 8.9% and 6.1%, respectively, when we include the control variables.

In columns (3) and (6) of Panel A, we examine the dynamic response of process innovation to the adoption of the good-faith exception. To this end, we replace *Good Faith* by *Good Faith -2*, *Good Faith -1*, *Good Faith 0*, *Good Faith +1*, *Good Faith +2*, and *Good Faith 3+*. These are indicator variables equal to one if the firm's state of headquarters will adopt the exception in two years, will adopt it the following year, adopted it in the current year, adopted it the year

before, adopted it two years before, and adopted it three or more years before, respectively. All control variables are included in these regressions. Consistent with the parallel-trends assumption required for a causal interpretation of our findings, we find no statistically or economically significant difference in the process innovation of firms in treated and control states in the years prior to the adoption of the good-faith exception (the coefficients of *Good Faith* -2 and *Good Faith* -1 are statistically insignificant). The increase in process innovation occurs three years following the adoption as time is required to shift the focus of firms' innovation efforts towards more process innovation.

In Panel B, the dependent variables are our measures of non-process innovation $\text{Log}(1 + \text{Non-Process Claims})$ in columns (1)-(3) and $\text{Log}(1 + \text{C-W Non-Process Patents})$ in columns (4)-(6). Columns (1)-(2) show that the adoption of the good-faith exception has no statistically or economically significant effect on non-process innovation measured using non-process claims. Column (4) shows a positive effect with marginal statistical significance in the specification without control variables when non-process innovation is measured using citation-weighted non-process patents, but this effect becomes both economically and statistically insignificant when we add the control variables in column (5). Further, the dynamic tests in columns (3) and (6) show that, for both measures of non-process innovation, there are no differences in the non-process innovation of firms in treated and control states in the years before or after the adoption of the good-faith exception. In sum, we find no meaningful impact of wrongful discharge laws on non-process innovation.

Overall, the results in Table 2 support the view that labor market rigidities that increase firms' operating leverage lead firms to tilt their innovation efforts toward developing new production processes. Note that, consistent with the discussion in Section 2.2 and prior evidence that the good-faith exception places the greatest rigidities on labor markets, we find no effect of the implied-contract or public-policy exceptions on firms' process or non-process innovation. Hence, while we include *Implied Contract* and *Public Policy* in all subsequent analyses, in the remainder of the paper we focus attention on the good-faith exception and the mechanism through which it affects innovation.

3.3 The effect of Good Faith on innovation and the role of the industry labor cost share

We next examine whether the effect of the adoption of the good-faith exception on innovation is related to the importance of labor costs in the industry's cost structure. Firms' incentives to invent new cost-saving production processes are likely to be greater in industries for which

labor costs account for a larger fraction of production costs, since in those industries labor costs are more important drivers of profitability and value.

To this end, Table 3 augments our main specification to include *LaborCostShr* and *Good Faith* $\times$ *LaborCostShr*, where *LaborCostShr* is the labor cost share of the firm’s 3-digit SIC industry. To facilitate the interpretation of the estimated coefficients, *LaborCostShr* is standardized to have mean zero and standard deviation of one. In all regression models, we include the same control variables as in Table 2. In these tests, we are able to also consider specifications that include state fixed effects interacted with year fixed effects, which absorb any unobservable state shocks that could confound our results.

The results in columns (1)-(4) show that the adoption of the good-faith exception leads to a larger increase in process innovation in industries for which labor costs account for a larger share of production costs: The coefficient of *Good Faith* $\times$ *LaborCostShr* is positive and statistically significant, both when we measure process innovation using process claims in columns (1)-(2) or citations-weighted process patents in columns (3)-(4). The estimated coefficients of *Good Faith* $\times$ *LaborCostShr* imply that, in industries whose *LaborCostShr* is one-standard deviation higher than the sample mean, the increase in process innovation is 5.9 to 6.7 percentage points larger when process innovation is measured by process claims and 5.3 to 5.9 percentage points larger when it is measured by citations-weighted process patents. In contrast, the results in columns (5)-(8) show that the recognition of the good-faith exception has no effect on non-process innovation, regardless of the share of labor costs in total costs in the firm’s industry.

3.4 Alternative explanations

In Table 4, we examine whether our main results could be due to unobserved changes in local economic conditions that possibly drive both the passage of wrongful discharge laws and the increase in process innovation. To this end, we augment of main specification in Table 2 to include *Good Faith Neighbor*, which is an indicator variable equal to one if a “neighboring state” (adjacent to the firm’s state of headquarters) has adopted the good-faith exception by year t and zero otherwise. If local economic conditions that are common for bordering states drive the results, then we should observe an increase in process innovation for firms located in states that do not adopt the legal change but share a common border with adopting states. However, the results in columns (1)-(2) show no effect of *Good Faith Neighbor* on a firm’s process innovation measured by process claims or by citations-weighted process patents,

while the coefficient of *Good Faith* remains positive and statistically significant. Columns (3)-(4) further show no effect of *Good Faith Neighbor* on a firm’s non-process innovation.

In Table 5, we address the concern that differential time trends in patent growth across states could bias our inferences. Lerner and Seru (2017) caution that a stronger enforcement of patent rights since the 1980s led to a surge in patenting activity thereafter that may have differentially affected innovation across states. Specifically, stronger patent rights could have had a differential impact across geographically-clustered fields of innovation, causing differential trends in innovation across states that could confound our inferences. To alleviate this concern, we estimate alternative specifications in which we control for these patterns in the data.

In Panel A, we focus on process innovation. In columns (1) and (4) we add the (lagged) logarithm of the mean number of patents filed by other firms located in the firm’s state *Log(State Patents)* to our baseline specification. The estimated effect of *Good Faith* on process innovation is very similar to that in the baseline results. In columns (2) and (5) we repeat the analysis over the 1975-1990 period, that is, the period before the explosive growth in patents observed post 1990, which occurred with different intensities across states. Despite the lower statistical power in a smaller sample, the coefficient of *Good Faith* remains statistically and economically significant. Last, in column (3) and (6) we exclude from the sample California and Massachusetts, which showed the highest patent growth over our sample period and account for a large share of patenting activity in the U.S. Again, the estimated coefficient of *Good Faith* remains positive and significant. Panel B repeats the same analyses using the two non-process innovation measures as the dependent variables and the coefficient of *Good Faith* remains statistically insignificant in all tests.

In Table 6, we investigate an alternative (but not exclusive) mechanism through which process innovation can increase following the adoption of the good-faith exception in a firm’s state of headquarters. The adoption of this exception in the firm’s state of headquarters increases job security for all of the firm’s workers employed in that state, including not only a firm’s production workers but also the inventors in charge of a firm’s innovation activities. It could therefore be that increased job security of a firm’s inventors employed at headquarters might incentivize them to exert more effort in innovation (Acharya, Baghai, and Subramanian (2014)) and such effort might be concentrated in developing new processes.

We analyze the text of each patent filing and extract the state of the lead inventor’s business address. This allows us to identify patents for which the lead inventor is employed

in a state other than the firm’s state of headquarters. For such inventors, their job security is generally unaffected by the adoption of the good-faith exception in the firm’s state of headquarters, because their employment contracts with the firm are typically signed under the labor laws of the state where they actually work. More generally, the job security argument applies to inventors that are directly employed by the firm, but does not apply to lead inventors working for the firm under temporary contracts but permanently employed in other institutions, for example, at universities, that are often located in other states.

Our tests are thus based on the importance of the location of a firm’s inventors. The effect of *Good Faith* on process innovation should be weaker when the firm’s inventors are employed in other states and thus unlikely to experience an increase in their job security. In contrast, the mechanism we highlight is that increased employment protection in a firm’s state of headquarters (where most of its production workers are located) incentivizes the firm to develop cost-saving process innovation. In this view, the state in which the firm’s inventors are located should be unrelated to the impact of *Good Faith* on process innovation.

In Panel A, we augment our main specification to include interactions of *Good Faith* with two indicator variables. The first is equal to one if more than 50% of the lead inventors listed in all patents the firm applied for in a given year are located outside the firm’s state of headquarters and zero otherwise (*Majority Inventors Outside*). The second is equal to one if no state concentrates more than 50% of the lead inventors listed in all patents the firm applied for in a given year and zero otherwise (*Disperse Inventors*). The estimated coefficients of *Good Faith* $\times$ *Majority Inventors Outside* and *Good Faith* $\times$ *Disperse Inventors* are statistically insignificant in all specifications we consider. For completeness, Panel B reports the results of analogous tests for non-process innovation. Again, the coefficients on both interactions are statistically insignificant. In sum, increased inventor job security is unlikely to be the driver of our results.

3.5 Stock of knowledge capital, adjustment of production methods, and firm performance

We now examine whether firms with greater available stock of knowledge capital are better able to adjust their production processes in response to increased labor protection and how such ability affects firm performance. Following Henderson and Cockburn (1996), we focus on a firm’s stock of prior patents to measure its knowledge capital.

In Table 7, we augment our regressions of process and non-process innovation on *Good Faith* to include an interaction of *Good Faith* with the firm’s lagged stock of patents $\text{Log}(1 +$

Patent Stock), which is standardized to a mean of zero and standard deviation of one. Columns (1)-(4) report the results for the two measures of process innovation and columns (5)-(8) report those for the two measures of non-process innovation.

We find a larger response of process innovation to the adoption of the good-faith exception for firms with a greater stock of knowledge capital: The estimated coefficient of *Good Faith* $\times$ *Log(1 + Patent Stock)* is positive and statistically significant in columns (1)-(4). These results imply that for firms with a one-standard deviation above the sample mean stock of patents, the adoption of the good-faith exception leads to an increase in process innovation that is 8.1% to 9.8% higher when measured with process claims and 15.1% to 15.5% when measured with citations-weighted process patents (relative to a firm with the stock of patents at the sample mean). Instead, the estimated coefficient of *Good Faith* $\times$ *Log(1 + Patent Stock)* is statistically insignificant in all specifications that use non-process innovation as the dependent variables (columns (5)-(8)). These results show that firms with a greater available stock of knowledge capital are better able to develop new process innovation in response to higher labor adjustment costs.

In Table 8, we examine whether firms with a greater stock of knowledge capital adjust their employment and capital intensity following the adoption of the good-faith exception. We consider four alternative dependent variables: the level of employment, *Log(Employment)*, in columns (1)-(2); the annual percentage change in employment, *Employment Growth*, in columns (3)-(4); the capital-labor ratio, *Log(K/L)*, in columns (5)-(6); and capital expenditures per employee, *Log(Capex/Emp)*, in columns (7)-(8). As in Table 7, we include *Good Faith* and *Good Faith* $\times$ *Log(1 + Patent Stock)* and all control variables in the regressions.

The coefficient of *Good Faith* is negative when the dependent variables are employment and employment growth, and positive when the dependent variables are the capital-labor ratio and capital expenditures per employee. The signs of these coefficients are consistent with the average firm reducing employment and substituting capital for labor, but the coefficients are statistically insignificant in all cases. Hence, we find no evidence of a major adjustment in employment or capital intensity for a firm with *Log(1 + Patent Stock)* at the sample mean. However, the coefficients of *Good Faith* $\times$ *Log(1 + Patent Stock)* are statistically significant and economically large for all four dependent variables. These results show that, as a result of higher labor adjustment costs, firms with a greater stock of knowledge capital not only increase their process innovation but they also adjust their production methods: They reduce employment, have lower employment growth, and they increase both their

capital-labor ratios and capital expenditures per employee. Thus, it is firms with a greater stock of knowledge capital that develop more process innovation and use it to implement new production methods that rely more on capital and less on labor.

In Table 9 we ask whether the adjustment in production methods permitted by a greater available stock of knowledge mitigates the negative impact of labor protection on firm performance. Specifically, we examine if the impact of the adoption of the good-faith exception on a firm's labor productivity and financial performance depends on its knowledge capital. The dependent variables are labor productivity measured by sales per employee, $\text{Log}(\text{Sales}/\text{Emp})$, operating performance measured by operating income before depreciation scaled by assets, Profit , and equity value measured by the market-to-book equity ratio, $\text{Log}(\text{ME}/\text{BE})$. The independent variables include *Good Faith* and $\text{Good Faith} \times \text{Log}(1 + \text{Patent Stock})$, as well as control variables. We consider specifications with firm fixed effects and year fixed effects and those with firm fixed effects and state fixed effects interacted with year fixed effects.

Column (1) shows that, for a firm with patent stock at the sample mean, the impact of *Good Faith* on labor productivity is positive. This is consistent with firms taking actions that increase the productivity of labor after an increase in labor adjustment costs. However, the coefficient is small and statistically insignificant. Importantly, in both columns (1) and (2), the coefficients of $\text{Good Faith} \times \text{Log}(1 + \text{Patent Stock})$ are positive, statistically significant, and economically large. The estimates in column (1) imply that a firm with a patent stock one-standard deviation above the sample mean increases its labor productivity by 6.9% after the adoption of the good-faith exception. In contrast, a firm with a patent stock one-standard deviation below the sample mean decreases its labor productivity by 6.9%. These results imply a 13.8 percentage point differential change in the labor productivity of firms with high and low patent stock that is statistically significant.⁸ The coefficient of $\text{Good Faith} \times \text{Log}(1 + \text{Patent Stock})$ is positive and statistically significant in the profitability specification with firm fixed effects and year fixed effects (column (3)). This result implies a statistically significant 1.2 percentage point differential change in the profitability of firms with high and low patent stock. The coefficient remains of similar magnitude but is statistically

⁸ Besides lower employment, labor productivity can increase for other reasons. The introduction of new production technologies can cause a shift in the composition of a firm's labor towards using more skilled labor (Autor and Katz (1999), Autor, Levy, and Murnane (2003), Autor and Dorn (2013)). Higher firing costs can lead firms to more carefully select new hires and ultimately increase the quality of its workforce.

insignificant when we use firm fixed effects and state fixed effects interacted with year fixed effects (column (4)).

Consistent with labor rigidity hurting value, the effect of the adoption of the good-faith exception on firm equity value is negative for a firm with a patent stock at the sample mean (although the coefficient of *Good Faith* is statistically insignificant). Importantly, in both columns (5) and (6), the coefficients of $Good\ Faith \times \log(1 + Patent\ Stock)$ are positive, statistically significant, and economically large. The estimates in column (5) imply that a firm with a patent stock one-standard deviation above the sample mean experiences a 1.9% increase in its equity value after the adoption of the good-faith exception. In contrast, a firm with a patent stock one-standard deviation below the sample mean experiences a 7.1% decrease in its equity value. The results imply a statistically significant 8.4 to 9.0 percentage point differential change in the equity value of firms with high and low patent stock.

The results in Table 9 suggest that, through the process innovation mechanism we describe, firms with a greater stock of knowledge capital are better able to boost the productivity of their labor and to maintain their profitability, ultimately avoiding economically significant value losses when increased labor rigidity hurts their business. They also highlight that a firm's stock of knowledge capital is a key determinant of its success in adjusting to changing conditions in input markets and, ultimately, a key driver of firm value.

3.6 Robustness and additional tests

We now discuss the results of additional tests that help assess the robustness of our main results in Table 2. These results are reported in tables A1-A2 of the Online Appendix.

First, we examine the robustness of our results based on citations-weighted process and non-process patents reported in columns (4)-(6) of Table 2 to the choice of the weighting scheme. In columns (1)-(2) and (5)-(6) of Table A1, we instead use raw (not weighted) process patents (*Process Patents*) and non-process patents (*Non-Process Patents*) as the dependent variables. These measures simply count the number of patents of each type and thus give equal weights to all patents regardless of their significance. In columns (3)-(4) and (7)-(8) we use market value-weighted counts of process patents (*V-W Process Patents*) and non-process patents (*V-W Non-Process Patents*) as the dependent variables. The market value of each patent comes from Kogan, Papanikolaou, Seru, and Stoffman (2017). These measures weight patents using the economic value of the innovation contained in the patent rather than the scientific merit gauged by citations.

The results in columns (1)-(4) continue to indicate a positive and statistically significant impact of the adoption of the good-faith exception on process innovation. Compared to the results based on citations-weighted patents reported in Table 2, the coefficient of *Good Faith* is smaller when the dependent variable is not weighted (columns (1)-(2)) and similar in magnitude when the dependent variable is value weighted (columns (3)-(4)). This suggests that there is heterogeneity in the value of the innovation contained in different patents, and that increased incentives to develop new process innovation lead to both a larger count of process patents and to more valuable process innovations contained in those patents. The results on non-process innovation in columns (5)-(8) are also consistent with those reported in Table 2. Once the control variables are included in the regressions, there is not statistically significant impact of the adoption of the good-faith exception on non-process innovation. Further, the coefficients of *Good Faith* are positive but smaller than for process innovation.

In Table A2, we restrict attention to firm-years in which firms filed at least one process or non-process patent and examine the impact of the adoption of the good-faith exception on the composition of patented innovation. We consider four alternative dependent variables: the share of process claims in the total number of claims contained in all patents filed by the firm (*Process Share in Claims*), the share of citations-weighted process patents in the total number of citation-weighted process and non-process patents filed by a firm (*Process Share in C-W Patents*), the share of (unweighted) process patents in the total number of (unweighted) process and non-process patents filed by a firm (*Process Share in Patents*), and the share of value-weighted process patents in the value-weighted number of process and non-process patents filed by a firm (*Process Share in V-W Patents*). These regressions include firm-fixed effects and year fixed effects and cluster the standard errors by state. The control variables are the same as in Table 2.

Across all specifications, we find that the adoption of the good-faith exception leads to a statistically significant increase in the share of process innovation. In terms of magnitudes, the estimated coefficients of *Good Faith* in the specifications with control variables indicate that the share of process innovation increases by 2.1 percentage points when the dependent variable is *Process Share in Claims* (or 8.6% relative to the sample mean for this variable) and by 3.0 percentage points when the dependent variable is *Process Share in C-W Patents* (or 15.1% relative to the sample mean for this variable). The results based on *Process Share in Patents* and *Process Share in V-W Patents* are very similar: The share of process innovation increases by 15.7% and 15.8% relative to the sample mean of these variables, respectively.

4. Conclusions

Our evidence highlights that increases in labor rigidity steer firms' innovation activities toward process innovation that facilitates the adjustment of production methods and thereby mitigates the negative effects of such rigidity on firm value. More broadly, our paper suggests that a greater available stock of knowledge capital allows firms to more easily adjust their strategies when business conditions change and that, through this mechanism, innovation is a driver of firms' performance.

From a public policy point of view, we highlight the unintended (or at least not fully understood) consequences of labor regulation that aims to benefit employees. Labor laws that increase employment protection or otherwise make labor costlier ultimately lead firms to take actions to mitigate the impact of these laws on their performance. This can negate some of the benefits of employment protection sought by the regulation. Author (2003) documents an increase in outsourcing. We highlight another mechanism, namely, a change in corporate R&D policy toward introducing production methods that are less reliant on labor inputs.

Our study also stresses the importance of distinguishing different types of innovation. Prior studies examine the impact of regulatory changes on overall innovation, but such changes do not have to affect all types of innovation in the same way. The new measures of process innovation we develop in this paper can thus aid future research that seeks to understand the determinants of innovation.

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Appendix: Variable Definitions

Process and non-process innovation measures

Process Claims	The number of process claims contained in all patents filed by the firm in each year.
Non-Process Claims	The number of non-process claims contained in all patents filed by the firm in each year.
C-W Process Patents	The citations-weighted number of process patents filed by the firm in each year. Citations received are counted, for each patent, over the 3-year window period after the patent award year. Process patents are patents that contain only process claims.
C-W Non-Process Patents	The citations-weighted number of non-process patents filed by the firm in each year. Citations received are counted, for each patent, over the 3-year window period after the patent award year. Non-Process patents are patents that contain only non-process claims.

Wrongful Discharge Laws indicators (coding as in Autor, Donohue, and Schwab (2006)):

Good Faith	Indicator variable equal to one if the firm's state of headquarters has adopted the Good Faith exception to the "firing at will" doctrine by year t , and zero otherwise.
Implied Contract	Indicator variable equal to one if the firm's state of headquarters has adopted the Implied Contract exception to the "firing at will" doctrine by year t , and zero otherwise.
Public Policy	Indicator variable equal to one if the firm's state of headquarters has adopted the Public Policy exception to the "firing at will" doctrine by year t , and zero otherwise.

Additional Good Faith variables

Good Faith -2	Indicator variable equal to one if the firm's state of headquarters will adopt the Good Faith exception in two years, and zero otherwise.
Good Faith -1	Indicator variable equal to one if the firm's state of headquarters will adopt the Good Faith exception in the following year, and zero otherwise.
Good Faith 0	Indicator variable equal to one if the firm's state of headquarters adopted the Good Faith exception in the current year, and zero otherwise.
Good Faith +1	Indicator variable equal to one if the firm's state of headquarters adopted the Good Faith exception the year before, and zero otherwise.
Good Faith +2	Indicator variable equal to one if the firm's state of headquarters adopted the Good Faith exception two years before, and zero otherwise.
Good Faith 3+	Indicator variable equal to one if the firm's state of headquarters adopted the Good Faith exception three or more years before, and zero otherwise.
Good Faith Neighbor	Indicator variable equal to one if a "neighboring state", that is, a state adjacent to the firm's state of headquarters, has adopted the Good Faith exception by year t , and zero otherwise.

Employment and capital intensity variables

Employment Growth	Annual growth rate of a firm's employment (<i>emp</i>).
Log(Employment)	Logarithm of a firm's employment level (<i>emp</i>).
Log(K/L)	Logarithm of a firm's capital-labor ratio defined as property, plant, and equipment (<i>ppent</i>) divided by employment (<i>emp</i>).
Log(Capex/Emp)	Logarithm of a firm's capital expenditures per employee defined as capital expenditures (<i>capex</i>) divided the employment (<i>emp</i>).

Performance and value measures

Log(Sales/Emp)	Logarithm of sales (<i>sale</i>) dividend by the number of employees (<i>emp</i>).
Profit	Operating income before depreciation (<i>oibdp</i>) divided by assets (<i>at</i>).
Log(ME/BE)	Logarithm of market equity (<i>cshe</i> $\times$ <i>prcc_f</i>) divided by book equity (<i>at - lt</i>).

Control variables

Patent Stock	A firm's patent stock computed by adding its patents since 1872 and assuming an annual depreciation rate of 15%.
R&D Stock	A firm's R&D stock computed by adding its R&D spending (<i>xrd</i>) since 1950 and assuming an annual depreciation rate of 15%.
Log(Sales)	The logarithm of a firm's sales (<i>sale</i>).
Log(M/B)	The logarithm of a firm's market value of assets (the sum of the market value of equity, <i>cshe</i> $\times$ <i>prcc_f</i> , the book value of long term debt, <i>dltt</i> , and the book value of debt in current liabilities, <i>dlc</i>) scaled by the book value of assets (<i>at</i>).
State GDP Growth	Annual growth rate of state GDP in current dollars (Source: U.S. Bureau of Economic Analysis).
State Political Balance	Fraction of a state's congress members in the U.S. House of Representatives that belong to the Democratic Party (Source: History, Art & Archives, U.S. House of Representatives).

Other variables

LaborCostShr	The average labor cost share in the firm's 3-digit SIC industry. For each firm, dollar labor costs are computed as the Compustat number of employee's (<i>emp</i>) times the average wage rate in the firm's industry obtained from the Quarterly Census of Employment and Wages provided by the U.S. Bureau of Labor Statistics. Next, the dollar labor costs are divided by the firm's cost of goods sold (<i>cogs</i>). Last, firm cost shares are averaged across all firm-years in each 3-digit SIC industry.
Log(State Patents)	The logarithm of the mean number of patents filed in each year by firms headquartered in the same state as the firm. Patents filed by the firm itself are excluded from the calculation of the mean.

Majority Inventors Outside	Indicator variable equal to one if more than 50% of the lead inventors listed in all patents the firm applied for in a given year are located outside the firm's state of headquarters, and zero otherwise. The lead inventor is the first inventor listed on the patent grant document.
Disperse Inventors	Indicator variable equal to one if no U.S. state concentrates more than 50% of the lead inventors listed in all patents the firm applied for in a given year, and zero otherwise. The lead inventor is the first inventor listed on the patent grant document.

Table 1: Summary Statistics

The sample spans the period 1975-1997. It includes all non-financial and non-utility firms headquartered in the U.S. that filed at least one patent with the USPTO during this period and for which we can compute the key variables. All variables are defined in the Appendix.

	Mean	Std. Dev.	Pctile 10	Median	Pctile 90
<i><u>Process and non-process innovation measures</u></i>					
Log(1 + Process Claims)	1.133	1.825	0.000	0.000	4.025
Log(1 + Non-Process Claims)	1.872	2.195	0.000	0.000	5.017
Log(1 + C-W Process Patents)	0.507	1.214	0.000	0.000	2.303
Log(1 + C-W Non-Process Patents)	1.126	1.658	0.000	0.000	3.664
<i><u>Wrongful Discharge Laws indicators</u></i>					
Good Faith	0.238	0.426	0.000	0.000	1.000
Implied Contract	0.638	0.480	0.000	1.000	1.000
Public Policy	0.654	0.476	0.000	1.000	1.000
<i><u>Employment and capital intensity variables</u></i>					
Employment growth	0.037	0.254	-0.189	0.023	0.288
Log(Employment)	0.283	2.054	-2.375	0.262	3.046
Log(K/L)	2.988	1.086	1.725	2.911	4.359
Log(Capex/Emp)	1.424	1.197	0.000	1.383	2.917
<i><u>Performance and value measures</u></i>					
Log(Sales/Emp)	4.543	0.793	3.665	4.536	5.461
Profit	0.099	0.190	-0.064	0.136	0.249
Log(ME/BE)	0.494	0.857	-0.497	0.423	1.553
<i><u>Control variables</u></i>					
Log(1 + Patent Stock)	1.708	1.609	0.000	1.314	3.984
Log(1 + R&D Stock)	1.830	1.928	0.000	1.302	4.623
Log(Sales)	4.691	2.264	1.930	4.687	7.617
Log(M/B)	0.116	0.715	-0.630	-0.018	1.085
State GDP Growth	0.077	0.036	0.035	0.075	0.124
State Political Balance	0.602	0.164	0.435	0.600	0.833

Table 2: Wrongful Discharge Laws and Process vs. Non-Process Innovation

The table reports the results of OLS regressions of process innovation and non-process innovation on *Good Faith*, the other Wrongful Discharge Laws indicators (*Implied Contract* and *Public Policy*), lagged controls variables (*Log(1+Patent Stock)*, *Log(1+R&D Stock)*, *Log(Sales)*, *Log(M/B)*, *State GDP Growth*, and *Political Balance*), as well as firm fixed effects and year fixed effects. In Panel A, the dependent variables are *Log(1+Process Claims)* and *Log(1+C-W Process Patents)*. In Panel B, the dependent variables are *Log(1+Non-Process Claims)* and *Log(1+C-W Non-Process Patents)*. The tests in columns (3) and (6) replace *Good Faith* by *Good Faith -2*, *Good Faith -1*, *Good Faith 0*, *Good Faith +1*, *Good Faith +2*, and *Good Faith 3+*, which are indicator variables equal to one if the firm's state of headquarters will adopt the exception in two years, will adopt it the following year, adopted it in the current year, adopted it the year before, adopted it two years before, or adopted it three or more years before, respectively. All other variables are defined in the Appendix. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Impact of Good Faith on process innovation

	Process Claims			C-W Process Patents		
	(1)	(2)	(3)	(4)	(5)	(6)
Good Faith	0.134*** (4.456)	0.089*** (2.959)		0.091*** (3.062)	0.061** (2.119)	
Good Faith -2			-0.060 (-1.478)			-0.028 (-0.821)
Good Faith -1			-0.004 (-0.069)			0.052 (1.211)
Good Faith 0			0.008 (0.183)			0.007 (0.280)
Good Faith +1			-0.028 (-0.580)			0.024 (0.687)
Good Faith +2			0.013 (0.307)			0.010 (0.284)
Good Faith 3+			0.111** (2.279)			0.081** (2.379)
Implied Contract	0.002 (0.030)	-0.022 (-0.561)	-0.013 (-0.354)	0.002 (0.054)	-0.016 (-0.626)	-0.012 (-0.566)
Public Policy	-0.022 (-0.509)	-0.002 (-0.062)	-0.005 (-0.172)	-0.030 (-0.939)	-0.018 (-0.811)	-0.024 (-1.113)
Log(1 + Patent Stock)		0.422*** (14.699)	0.421*** (14.590)		0.281*** (7.876)	0.282*** (7.929)
Log(1 + R&D Stock)		0.178*** (6.090)	0.176*** (6.097)		0.114*** (5.849)	0.113*** (5.881)
Log(Sales)		0.105*** (6.785)	0.107*** (6.972)		0.037*** (2.930)	0.037*** (2.922)
Log(M/B)		0.074*** (4.454)	0.072*** (4.302)		0.045*** (4.750)	0.046*** (4.839)
State GDP Growth		-0.347 (-1.026)	-0.348 (-1.161)		-0.387 (-1.461)	-0.378 (-1.498)
State Political Balance		-0.112 (-1.353)	-0.113 (-1.383)		-0.042 (-0.690)	-0.043 (-0.696)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	44,898	44,898	44,606	44,898	44,898	44,606
Adjusted R ²	0.706	0.728	0.729	0.713	0.733	0.733

Panel B: Impact of Good Faith on non-process innovation

	Non-Process Claims			C-W Non-Process Patents		
	(1)	(2)	(3)	(4)	(5)	(6)
Good Faith	0.070 (1.488)	0.027 (0.567)		0.054* (1.758)	0.021 (0.789)	
Good Faith -2			-0.090 (-1.156)			-0.035 (-1.009)
Good Faith -1			-0.084 (-1.309)			-0.021 (-0.550)
Good Faith 0			-0.118 (-1.171)			-0.048 (-0.935)
Good Faith +1			-0.092 (-1.197)			-0.018 (-0.482)
Good Faith +2			-0.047 (-0.696)			-0.014 (-0.332)
Good Faith 3+			-0.013 (-0.208)			0.003 (0.069)
Implied Contract	0.007 (0.135)	-0.012 (-0.256)	-0.006 (-0.141)	-0.003 (-0.073)	-0.018 (-0.586)	-0.014 (-0.458)
Public Policy	-0.002 (-0.039)	0.019 (0.478)	0.016 (0.392)	-0.005 (-0.130)	0.010 (0.284)	0.008 (0.225)
Log(1 + Patent Stock)		0.416*** (11.246)	0.410*** (10.803)		0.336*** (11.644)	0.332*** (11.278)
Log(1 + R&D Stock)		0.142*** (4.086)	0.141*** (4.086)		0.106*** (4.506)	0.106*** (4.511)
Log(Sales)		0.189*** (9.653)	0.190*** (9.558)		0.120*** (8.271)	0.120*** (8.204)
Log(M/B)		0.107*** (5.274)	0.103*** (5.191)		0.094*** (6.080)	0.092*** (6.098)
State GDP Growth		-0.407 (-0.921)	-0.443 (-1.076)		-0.175 (-0.594)	-0.218 (-0.767)
State Political Balance		-0.067 (-0.917)	-0.079 (-1.134)		-0.022 (-0.351)	-0.018 (-0.283)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	44,898	44,898	44,606	44,898	44,898	44,606
Adjusted R ²	0.677	0.694	0.695	0.715	0.733	0.733

Table 3: Cross-Sectional Variation in the Effect of Good Faith on Innovation – The Role of the Industry Labor Cost Share

The table reports the results of OLS regressions of process and non-process innovation on *Good Faith* that further interact *Good Faith* with the average labor cost share in the firm's 3-digit SIC industry, *LaborCostShr* (standardized to a mean of zero and standard deviation of one). The dependent variables are *Log(1+Process Claims)* in columns (1)-(2), *Log(1+C-W Process Patents)* in columns (3)-(4), *Log(1+Non-Process Claims)* in columns (5)-(6), and *Log(1+C-W Non-Process Patents)* in columns (7)-(8). All regressions include *Implied Contract* and *Public Policy* and the lagged controls variables (*Log(1+Patent Stock)*, *Log(1+R&D Stock)*, *Log(Sales)*, *Log(M/B)*, *State GDP Growth*, and *Political Balance*). All variables are defined in the Appendix. Specifications (1), (3), (5), and (7) include firm fixed effects and year fixed effects. Specifications (2), (4), (6), and (8) include firm fixed effects and state fixed effects interacted with year fixed effects. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Process Innovation				Non-Process Innovation			
	Process Claims		C-W Process Patents		Non-Process Claims		C-W Non-Process Patents	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Good Faith	0.095*** (2.829)		0.066** (2.116)		0.030 (0.641)		0.022 (0.823)	
Good Faith × LaborCostShr	0.059* (1.777)	0.067** (2.056)	0.053* (1.962)	0.059** (2.328)	0.013 (0.328)	0.017 (0.385)	-0.001 (-0.036)	0.001 (0.028)
LaborCostShr	0.047 (1.372)	0.048 (1.280)	0.019 (0.820)	0.025 (1.007)	0.107*** (3.383)	0.109*** (3.797)	0.071*** (2.725)	0.078*** (3.113)
Other WDLs & Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	No	Yes	No	Yes	No	Yes	No
State × Year Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Observations	44,860	44,765	44,860	44,765	44,860	44,765	44,860	44,765
Adjusted R ²	0.728	0.729	0.733	0.733	0.694	0.695	0.733	0.733

Table 4: Adoption of the Good Faith Exemption in Neighboring States

The table reports the results of OLS regressions of process and non-process innovation on *Good Faith* and an indicator variable *Good Faith Neighbor* for whether a “neighboring state” (a state adjacent to the firm’s state of headquarters) has adopted the Good Faith exception by year t and zero otherwise. The dependent variables are $\text{Log}(1+\text{Process Claims})$ in column (1), $\text{Log}(1+\text{C-W Process Patents})$ in column (2), $\text{Log}(1+\text{Non-Process Claims})$ in column (3), and $\text{Log}(1+\text{C-W Non-Process Patents})$ in column (4). All regressions include *Implied Contract* and *Public Policy*, the lagged controls variables ($\text{Log}(1+\text{Patent Stock})$, $\text{Log}(1+\text{R\&D Stock})$, $\text{Log}(\text{Sales})$, $\text{Log}(M/B)$, *State GDP Growth*, and *Political Balance*), as well as firm-fixed effects and year fixed effects. All variables are defined in the Appendix. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Process Innovation		Non-Process Innovation	
	Process Claims (1)	C-W Process Patents (2)	Non-Process Claims (3)	C-W Non-Process Patents (4)
Good Faith	0.086*** (2.999)	0.063** (2.208)	0.022 (0.465)	0.016 (0.628)
Good Faith Neighbor	-0.021 (-0.555)	0.013 (0.729)	-0.039 (-0.946)	-0.040 (-1.475)
Other WDLs & Controls	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	44,898	44,898	44,898	44,898
Adjusted R ²	0.728	0.733	0.694	0.733

Table 5: Controlling for Time Trends in Innovation

The table reports the results of OLS regressions of process innovation and non-process innovation on *Good Faith*, the other Wrongful Discharge Laws indicators (*Implied Contract* and *Public Policy*), lagged controls variables (*Log(1+Patent Stock)*, *Log(1+R&D Stock)*, *Log(Sales)*, *Log(M/B)*, *State GDP Growth*, and *Political Balance*), as well as firm fixed effects and year fixed effects. In Panel A, the dependent variables are *Log(1+Process Claims)* in columns (1)-(3) and *Log(1+C-W Process Patents)* in columns (4)-(6). In Panel B, the dependent variables are *Log(1+Non-Process Claims)* in columns (1)-(3) and *Log(1+C-W Non-Process Patents)* in columns (4)-(6). For both panels, columns (1) and (4) include *Log(State Patents)*, defined as the logarithm of the mean number of patents filed by firms headquartered in the firm's state in each year (the patents filed by the firm itself are excluded in the calculation of this mean), as an additional control variable. For both panels, columns (2) and (4) estimate the benchmark specification restricting the sample to the period 1975-1990, and columns (3) and (6) estimate the benchmark specification excluding the states of California and Massachusetts from the sample. All variables are defined in the Appendix. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Effect of Good Faith on process innovation

	Process Claims			C-W Process Patents		
	(1)	(2)	(3)	(4)	(5)	(6)
Good Faith	0.080*** (2.726)	0.081*** (2.704)	0.122*** (4.920)	0.056** (2.142)	0.052* (1.964)	0.068** (2.056)
Log(State Patents)	0.091** (2.309)			0.050 (1.620)		
Sample	Full	1975-90	Excl. CA & MA	Full	1975-90	Excl. CA & MA
Other WDLs & Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	44,898	29,328	35,326	44,898	29,328	35,326
Adjusted R ²	0.728	0.756	0.740	0.733	0.786	0.761

Panel B: Effect of Good Faith on non-process innovation

	Non-Process Claims			C-W Non-Process Patents		
	(1)	(2)	(3)	(4)	(5)	(6)
Good Faith	0.015 (0.299)	-0.009 (-0.184)	0.030 (0.551)	0.013 (0.514)	-0.008 (-0.342)	-0.004 (-0.109)
Log(State Patents)	0.118** (2.018)			0.077* (2.008)		
Sample	Full	1975-90	Excl. CA & MA	Full	1975-90	Excl. CA & MA
Other WDLs & Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	44,898	29,328	35,326	44,898	29,328	35,326
Adjusted R ²	0.694	0.718	0.702	0.733	0.766	0.745

Table 6: Effect of Good Faith on Process vs. Non-Process Innovation – Lead Inventor’s State of Location

The table reports the results of OLS regressions of process and non-process innovation on *Good Faith* that further interact *Good Faith* with two variables that capture the location of inventors relative to the location of the firm. The first is an indicator variable equal to one if more than 50% of the lead inventors listed in all patents the firm applied for in a given year are located outside the firm’s state of headquarters and zero otherwise (*Majority Inventors Outside*). The second is an indicator variable equal to one if no state concentrates more than 50% of the lead inventors listed in all patents the firm applied for in a given year and zero otherwise (*Disperse Inventors*). In Panel A, the dependent variables are *Log(1+Process Claims)* in columns (1)-(4) and *Log(1+C-W Process Patents)* in columns (5)-(8). In Panel B, the dependent variables are *Log(1+Non-Process Claims)* in columns (1)-(4) and *Log(1+C-W Non-Process Patents)* in columns (5)-(8). All regressions include *Implied Contract* and *Public Policy* and the lagged controls variables (*Log(1+Patent Stock)*, *Log(1+R&D Stock)*, *Log(Sales)*, *Log(M/B)*, *State GDP Growth*, and *Political Balance*). All variables are defined in the Appendix. Specifications (1), (3), (5), and (7) include firm fixed effects and year fixed effects. Specifications (2), (4), (6), and (8) include firm fixed effects and state fixed effects interacted with year fixed effects. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Regressions of process innovation on Good Faith interacted with Majority Inventors Outside and Disperse Inventors

	Process Claims				C-W Process Patents			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Good Faith	0.084*** (2.896)		0.073** (2.339)		0.055* (1.891)		0.053* (1.966)	
Good Faith × Majority Inventors Outside	0.047 (0.501)	0.030 (0.345)			0.084 (0.989)	0.073 (0.826)		
Majority Inventors Outside	0.664*** (12.269)	0.667*** (12.174)			0.288*** (5.218)	0.280*** (4.963)		
Good Faith × Disperse Inventors			0.146 (1.605)	0.135 (1.544)			0.078 (0.827)	0.071 (0.755)
Disperse Inventors			0.435*** (12.947)	0.435*** (13.028)			0.075*** (3.319)	0.073*** (3.241)
Other WDLs & Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	No	Yes	No	Yes	No	Yes	No
State × Year Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Observations	44,898	44,803	44,898	44,803	44,898	44,803	44,898	44,803
Adjusted R ²	0.732	0.733	0.731	0.732	0.735	0.735	0.733	0.734

Panel B: Regressions of non-process innovation on Good Faith interacted with Majority Inventors Outside and Disperse Inventors

	Non-Process Claims				C-W Non-Process Patents			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Good Faith	0.027 (0.525)		0.025 (0.456)		0.020 (0.728)		0.018 (0.619)	
Good Faith × Majority Inventors Outside	-0.026 (-0.286)	-0.026 (-0.319)			-0.003 (-0.037)	-0.000 (-0.005)		
Majority Inventors Outside	0.901*** (18.721)	0.898*** (18.568)			0.658*** (15.370)	0.649*** (15.321)		
Good Faith × Disperse Inventors			0.012 (0.181)	-0.004 (-0.066)			0.020 (0.256)	0.008 (0.107)
Disperse Inventors			0.890*** (23.760)	0.894*** (23.904)			0.515*** (15.690)	0.518*** (16.183)
Other WDLs & Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	No	Yes	No	Yes	No	Yes	No
State × Year Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Observations	44,898	44,803	44,898	44,803	44,898	44,803	44,898	44,803
Adjusted R ²	0.699	0.700	0.702	0.703	0.737	0.738	0.737	0.738

Table 7: Stock of Knowledge Capital and the Effect of Good Faith on Process vs. Non-Process Innovation

The table reports OLS regressions of process and non-process innovation on *Good Faith*, including an interaction of *Good Faith* with the firm's lagged stock of patents, *Log(1+Patent Stock)* (standardized to a mean of zero and standard deviation of one). The dependent variables are *Log(1+Process Claims)* in columns (1)-(2), *Log(1+C-W Process Patents)* in columns (3)-(4), *Log(1+Non-Process Claims)* in columns (5)-(6), and *Log(1+C-W Non-Process Patents)* in columns (7)-(8). All regressions include *Implied Contract* and *Public Policy* and the lagged controls variables (*Log(1+Patent Stock)*, *Log(1+R&D Stock)*, *Log(Sales)*, *Log(M/B)*, *State GDP Growth*, and *Political Balance*). All variables are defined in the Appendix. Specifications (1), (3), (5), and (7) include firm fixed effects and year fixed effects. Specifications (2), (4), (6), and (8) include firm fixed effects and state fixed effects interacted with year fixed effects. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Process Innovation				Non-Process Innovation			
	Process Claims		C-W Process Patents		Non-Process Claims		C-W Non-Process Patents	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Good Faith	0.065** (2.348)		0.024 (0.983)		0.028 (0.473)		0.012 (0.447)	
Good Faith × Log(1 + Patent Stock)	0.098* (1.931)	0.081* (1.808)	0.155** (2.519)	0.151** (2.522)	-0.002 (-0.034)	-0.023 (-0.422)	0.038 (0.759)	0.029 (0.659)
Log(1 + Patent Stock)	0.649*** (16.750)	0.647*** (15.971)	0.405*** (12.380)	0.408*** (12.419)	0.668*** (12.522)	0.661*** (11.304)	0.528*** (14.027)	0.524*** (12.609)
Other WDLs & Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	No	Yes	No	Yes	No	Yes	No
State × Year Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Observations	44,898	44,803	44,898	44,803	44,898	44,803	44,898	44,803
Adjusted R ²	0.728	0.729	0.734	0.734	0.694	0.695	0.733	0.733

Table 8: Stock of Knowledge Capital and the Effect of Good Faith on Employment and Capital Intensity

The table reports OLS regressions of firm employment and capital intensity on *Good Faith*, including an interaction of *Good Faith* with the firm's lagged stock of patents, $\text{Log}(1+\text{Patent Stock})$ (standardized to a mean of zero and standard deviation of one). The regressions also include the other Wrongful Discharge Laws indicators (*Implied Contract* and *Public Policy*) and lagged control variables. The dependent variables are a firm's annual employment growth rate (*Employment Growth*) in columns (1)-(2), the logarithm of a firm's employment level ($\text{Log}(\text{Employment})$) in columns (3)-(4), the logarithm of a firm's capital-labor ratio defined as property, plant, and equipment divided by employment ($\text{Log}(K/L)$) in columns (5)-(6), and the logarithm of a firm's capital expenditures per employee ($\text{Log}(\text{Capex}/\text{Emp})$) in columns (7)-(8). All other variables are defined in the Appendix. Specifications (1), (3), (5), and (7) include firm fixed effects and year fixed effects. Specifications (2), (4), (6), and (8) include firm fixed effects and state fixed effects interacted with year fixed effects. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Employment Growth		Log(Employment)		Log(K/L)		Log(Capex/Emp)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Good Faith	-0.005 (-0.791)		-0.022 (-0.821)		0.030 (1.560)		0.016 (0.826)	
Good Faith $\times$ Log(1 + Patent Stock)	-0.010** (-2.083)	-0.009*** (-3.480)	-0.079*** (-6.390)	-0.067*** (-4.411)	0.067*** (6.091)	0.074*** (3.824)	0.091*** (3.764)	0.101** (2.553)
Log(1 + Patent Stock)	0.000 (0.007)	-0.000 (-0.095)	0.110*** (9.668)	0.109*** (9.004)	0.042*** (2.907)	0.041*** (2.825)	-0.038* (-1.721)	-0.040* (-1.713)
Implied Contract	-0.004 (-0.653)		-0.045** (-2.259)		-0.014 (-0.878)		0.002 (0.108)	
Public Policy	-0.006 (-1.126)		-0.008 (-0.472)		-0.035 (-1.638)		-0.038 (-1.443)	
Log(1 + R&D Stock)	0.087*** (15.139)	0.087*** (15.028)	0.085*** (10.455)	0.082*** (9.978)	0.049*** (4.209)	0.049*** (4.056)	0.385*** (20.882)	0.379*** (19.584)
Log(Sales)	-0.068*** (-19.109)	-0.070*** (-19.706)	0.536*** (22.121)	0.529*** (21.784)	0.066*** (6.458)	0.064*** (6.042)	0.106*** (6.486)	0.104*** (6.059)
Log(M/B)	-0.024*** (-4.600)	-0.024*** (-4.399)	0.041** (2.609)	0.044*** (2.785)	0.035*** (3.471)	0.037*** (3.978)	-0.024* (-1.879)	-0.020 (-1.557)
State GDP Growth	0.115* (1.864)		0.422*** (4.812)		-0.221 (-1.311)		0.964*** (3.538)	
State Political Balance	-0.010 (-0.694)		-0.032 (-0.869)		0.025 (0.503)		0.105** (2.255)	
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	No	Yes	No	Yes	No	Yes	No
State $\times$ Year Fixed Effects	No	Yes	No	Yes	No	Yes	No	Yes
Observations	43,935	43,840	44,216	44,120	44,166	44,070	43,567	43,465
Adjusted R ²	0.175	0.180	0.967	0.967	0.859	0.861	0.697	0.699

Table 9: Stock of Knowledge Capital and the Effect of Good Faith on Firm Performance and Value

The table reports OLS regressions of firm performance and value measures on *Good Faith*, including an interaction of *Good Faith* with the firm's lagged stock of patents, *Log(1+Patent Stock)* (standardized to a mean of zero and standard deviation of one). The regressions also include the other Wrongful Discharge Laws indicators (*Implied Contract* and *Public Policy*) and lagged control variables. All variables are defined in the Appendix. The dependent variables are the logarithm of sales per employee (*Log(Sales/Emp)*) in columns (1)-(2), operating income before depreciation scaled by assets (*Profit*) in columns (3)-(4), and the logarithm of the market to book equity ratio (*Log(ME/BE)*) in columns (5)-(6). Specifications (1), (3), and (5) include firm fixed effects and year fixed effects. Specifications (2), (4), and (6) include firm fixed effects and state fixed effects interacted with year fixed effects. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Log(Sales/Emp)		Profit		Log(ME/BE)	
	(1)	(2)	(3)	(4)	(5)	(6)
Good Faith	0.022		0.001		-0.026	
	(1.270)		(0.130)		(-0.632)	
Good Faith × Log(1 + Patent Stock)	0.069***	0.059***	0.006***	0.005	0.045**	0.042**
	(7.230)	(4.697)	(3.247)	(1.634)	(2.228)	(2.267)
Log(1 + Patent Stock)	-0.052***	-0.050***	-0.012***	-0.010***	-0.047**	-0.045**
	(-5.201)	(-4.647)	(-3.563)	(-3.058)	(-2.226)	(-2.130)
Implied Contract	0.028*		0.002		0.014	
	(1.754)		(0.564)		(0.441)	
Public Policy	-0.018		0.003		0.014	
	(-1.526)		(0.907)		(0.389)	
Log(Sales)	0.189***	0.189***	0.036***	0.036***	-0.130***	-0.130***
	(16.979)	(16.411)	(8.485)	(8.483)	(-9.991)	(-9.896)
Log(1 + R&D Stock)	-0.010	-0.012	-0.021***	-0.020***	-0.076***	-0.073***
	(-0.995)	(-1.140)	(-5.803)	(-5.372)	(-5.835)	(-5.559)
State GDP Growth	-0.078		0.097**		1.319***	
	(-0.918)		(2.129)		(6.125)	
State Political Balance	0.014		0.009		0.058	
	(0.480)		(0.940)		(0.816)	
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	No	Yes	No	Yes	No
State × Year Fixed Effects	No	Yes	No	Yes	No	Yes
Observations	44,118	44,021	44,744	44,648	43,853	43,761
Adjusted R ²	0.825	0.827	0.671	0.674	0.618	0.627
<u>Impact of <i>Good Faith</i> with <i>Log(1 + Patent Stock)</i> evaluated at:</u>						
One-Std. Dev. above mean	0.091***		0.007		0.019	
	(5.68)		(1.45)		(0.46)	
One-Std. Dev. below mean	-0.047**		-0.005		-0.071	
	(-2.08)		(-0.94)		(-1.46)	
Difference	0.138***	0.119***	0.012***	0.009	0.090**	0.084**
	(7.23)	(4.70)	(3.25)	(1.63)	(2.23)	(2.27)

Online Appendix to
**“Shielding firm value: Employment protection and
process innovation”**

Jan Bena, Hernán Ortiz-Molina, and Elena Simintzi

Historical state of headquarters

To merge the wrongful discharge laws indicators and other state level variables into our dataset of publicly traded innovating firms, we rely on a firm's historical state of headquarters. To this end, we use the "company header history" file in the legacy CRSP/Compustat Merged database available through Wharton Research Data Services. This file provides, for each firm, the time frame for each verified historical state of headquarters and allows us to accurately track changes in the headquarter state over time. When this information is missing for a particular year, we use the firm's next or last verified state of headquarters if available. In this way, we obtain the historical state of headquarters for 75% of the observations in our sample. The remaining 25% of observations correspond to firms with no data available in the company header history file. Most of these firms have the last reporting year in Compustat between 1975 and 1990 overlapping especially with the early part of our sample. For these firms, we use the state of headquarters reported in the "company header file", which is the state of headquarters as of the firm's last reporting year. For these firms, the headquarter state we use is close in time to most of their observations. For example, if a firm last appears in Compustat in 1985, the company header file records the state of headquarters in 1985 and we use this state in all prior years as well.

Table A1: Robustness of Main Results to Using Alternative Measures of Innovation

The table reports the results of OLS regressions of process and non-process innovation on *Good Faith* that also include the other Wrongful Discharge Laws indicators (*Implied Contract* and *Public Policy*) and lagged control variables (*Log(1+Patent Stock)*, *Log(1+R&D Stock)*, *Log(Sales)*, *Log(M/B)*, *State GDP Growth*, and *Political Balance*), as well as firm fixed effects and year fixed effects. The dependent variables are *Log(1+Process Patents)* in columns (1)-(2), *Log(1+V-W Process Patents)* in columns (3)-(4), *Log(1+Non-Process Patents)* in columns (5)-(6), and *Log(1+V-W Non-Process Patents)* in columns (7)-(8). *Log(1+Process Patents)* and *Log(1+Non-Process Patents)* are based on (unweighted) counts of process and non-process patents, respectively. *Log(1+V-W Process Patents)* and *Log(1+V-W Non-Process Patents)* are based on value-weighted counts of process and non-process patents, respectively. The value of each patent comes from Kogan, Papanikolaou, Seru, and Stoffman (2017). All other variables are defined in the Appendix. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Process Innovation				Non-Process Innovation			
	Process Patents		V-W Process Patents		Non-Process Patents		V-W Non-Process Patents	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Good Faith	0.054*** (2.752)	0.040** (2.074)	0.087*** (3.099)	0.059** (2.310)	0.049* (1.945)	0.032 (1.354)	0.056** (2.172)	0.026 (0.997)
Implied Contract	0.001 (0.047)	-0.008 (-0.559)	0.004 (0.131)	-0.011 (-0.482)	0.002 (0.085)	-0.005 (-0.237)	0.003 (0.092)	-0.009 (-0.318)
Public Policy	-0.016 (-0.875)	-0.010 (-0.737)	-0.018 (-0.553)	-0.007 (-0.287)	0.003 (0.117)	0.012 (0.548)	0.038 (1.001)	0.053* (1.933)
Log(1 + Patent Stock)		0.160*** (8.123)		0.250*** (5.849)		0.215*** (10.099)		0.283*** (8.615)
Log(1 + R&D Stock)		0.045*** (4.082)		0.109*** (4.964)		0.039** (2.299)		0.113*** (4.367)
Log(Sales)		0.029*** (4.953)		0.053*** (3.794)		0.084*** (8.259)		0.119*** (7.057)
Log(M/B)		0.020*** (3.819)		0.089*** (8.677)		0.044*** (4.799)		0.161*** (9.211)
State GDP Growth		-0.189 (-1.562)		-0.247 (-1.166)		-0.106 (-0.583)		-0.150 (-0.579)
State Political Balance		-0.032 (-1.017)		-0.042 (-0.779)		-0.031 (-0.733)		-0.073 (-1.290)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	44,898	44,898	44,898	44,898	44,898	44,898	44,898	44,898
Adjusted R ²	0.803	0.818	0.788	0.803	0.805	0.820	0.825	0.839

Table A2: Wrongful Discharge Laws and Share of Process Innovation

The table reports the results of OLS regressions of the share of process innovation in total innovation (conditional on a firm's filing at least one patent in that year) on *Good Faith* that also include the other Wrongful Discharge Laws indicators (*Implied Contract* and *Public Policy*) and lagged control variables (*Log(1+Patent Stock)*, *Log(1+R&D Stock)*, *Log(Sales)*, *Log(M/B)*, *State GDP Growth*, and *Political Balance*), as well as firm fixed effects and year fixed effects. In columns (1)-(2), the dependent variable is the share of process claims in the total number of claims contained in all patents filed by a firm in each year (*Process Share in Claims*). In columns (3)-(4), the dependent variable is the ratio of the citation-weighted number of process patents to the citation-weighted number of process and non-process patents filed by a firm in each year (*Process Share in C-W Patents*). In columns (5)-(6), the dependent variable is the share of (unweighted) process patents in the total number of (unweighted) process and non-process patents filed by a firm in each year (*Process Share in Patents*). In columns (7)-(8), the dependent variable is the ratio of the value-weighted number of process patents to the value-weighted number of process and non-process patents filed by a firm in each year (*Process Share in V-W Patents*). The value of each patent comes from Kogan, Papanikolaou, Seru, and Stoffman (2017). All other variables are defined in the Appendix. The standard errors are adjusted for heteroscedasticity and clustering at the state level (t-statistics in parentheses). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Process Share in Claims		Process Share in C-W Patents		Process Share in Patents		Process Share in V-W Patents	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Good Faith	0.023*	0.021*	0.034**	0.030**	0.028***	0.026**	0.029**	0.026**
	(2.003)	(1.932)	(2.433)	(2.301)	(2.903)	(2.555)	(2.444)	(2.286)
Implied Contract	0.006	0.005	0.011	0.007	0.011	0.010	0.008	0.005
	(1.040)	(0.941)	(1.227)	(0.956)	(1.547)	(1.459)	(1.121)	(0.855)
Public Policy	-0.011	-0.010	-0.009	-0.008	-0.013*	-0.013*	-0.010	-0.009
	(-1.332)	(-1.370)	(-1.076)	(-1.122)	(-1.799)	(-1.863)	(-1.187)	(-1.242)
Log(1 + Patent Stock)		0.005		0.010***		0.011**		0.008**
		(1.419)		(2.924)		(2.648)		(2.569)
Log(1 + R&D Stock)		0.012**		0.018***		0.013**		0.016***
		(2.578)		(3.382)		(2.160)		(3.101)
Log(Sales)		-0.004		-0.006		-0.002		-0.004
		(-0.920)		(-0.992)		(-0.379)		(-0.762)
Log(M/B)		0.001		0.002		0.003		0.004
		(0.122)		(0.387)		(0.666)		(0.829)
State GDP Growth		-0.033		-0.221**		-0.015		-0.173***
		(-0.492)		(-2.608)		(-0.175)		(-2.840)
State Political Balance		-0.017		-0.018		-0.005		-0.020
		(-0.727)		(-0.770)		(-0.208)		(-1.040)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,232	22,232	19,621	19,621	22,002	22,002	19,598	19,598
Adjusted R ²	0.414	0.415	0.411	0.412	0.357	0.358	0.434	0.435